

MUHAMMAD HAIDER ZAIDI

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Technical Skills

Programming: Python, C++, SQL, Javascript

Deep Learning: PyTorch, TensorFlow, CNNs, YOLO, LLMs, Transformer Models, Signal Processing, Few-Shot Learning

MLOps: Docker, Kubernetes, AWS SageMaker, FastAPI, TensorRT, Real-time System Optimization

Data Engineering: PySpark, Airflow, ETL Pipelines, Data Processing, Hadoop Ecosystem (Hive, Sqoop, NiFi)

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for metrological applications, significantly improving **model performance** and training efficiency through **digitalization** efforts.
- Explored novel architectures and **physics-informed hybrid loss functions** to enhance **model accuracy** and stability, contributing to robust system development.
- Collaborated with PhD researchers on advanced **Machine Learning** research, contributing to a publication at FLOMEKO conference (under review).
- Developed **dashboards for data visualisation** and **model performance monitoring**, enabling effective project oversight and data-driven decision making.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Enhanced **AI algorithm** performance for vitiligo segmentation, achieving 94% accuracy and 5x speed improvement, with deployment via **Docker** and **FastAPI** for efficient **real-time systems**.
- Engineered a **real-time** tennis foul detection system leveraging **TensorRT**, reducing compute cost and processing time by 6x, showcasing **system optimization** skills.
- Managed end-to-end **MLOps pipelines** on **AWS SageMaker**, effectively addressing **YOLO model data drift** and maintaining **Kubernetes infrastructure** for scalable **Machine Learning** solutions.
- Developed **Proof-of-Concepts (POCs)** using cutting-edge **YOLO**, **few-shot learning**, and **CNNs** for client demonstrations, translating complex **AI** concepts into tangible applications.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Executed full-stack development for a website offering **AI-based services**, utilizing Flask, HTML, and CSS, contributing to **digitalization** initiatives.
- Developed an **AI chatbot application** by integrating the OpenAI API and relevant **LLM tools/frameworks**, demonstrating practical application of advanced **Machine Learning**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented **ETL pipelines** for both structured and unstructured data, ensuring robust **data processing** for **Machine Learning** applications.
- Automated **data ingestion and cleaning workflows**, enhancing efficiency and reliability of data for client analytics and **digitalization** projects.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a novel **weighted-probability ensemble** combining DPFunc and DeepGOPlus, leveraging **AlphaFold** for **feature extraction** and **taxonomic bias mitigation** to advance **Machine Learning** in bioinformatics.
- Achieved a 4% increase in **model accuracy** through custom **feature engineering** across MF, CC, and BP namespaces, demonstrating proficiency in optimizing **deep learning models**.

Razanalytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a **real-time Bitcoin price forecasting system** as a thesis project, applying **deep learning models** (e.g., LSTMs, Transformers) for predictive analytics.
- Implemented robust **data pipelines** for continuous ingestion and processing of market data, crucial for accurate **real-time predictions** and **digitalization** of financial insights.

Saaz: AI Self-Playing Chitrati Sitar

2023

Signal Processing, Music Generation

- Designed real-time signal processing system integrating temporal dependencies for music generation
- Implemented transformer-based models for sequence prediction in audio generation tasks

Hugging Face

Open-Source

Contributor

- Co-authored the **Zero-shot Image Classification technical documentation** and implementation page on the Hugging Face Hub, contributing to the open-source **Machine Learning** community.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in **deep learning** for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in **Machine Learning** and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python